Lecture Transcript by Slide

Week 1 · Appendix A: Cardiac Anatomy and Physiology · 22 min

Auto-transcribed, so expect occasional errors, especially in technical terms. Timestamps are positions in the recording.

0:00:00 Appendix A: Cardiac Anatomy and Physiology (title slide)

0:00:35 Location of the Heart

0:01:23 Layers of Heart Wall

0:03:10 Internal Anatomy of the Heart

0:07:06 Walls of the Heart

0:07:32 Blood Flow through the Body

0:08:16 Heart Valves

0:10:42 Heart Valves (diagram)

0:10:59 Systole and Diastole

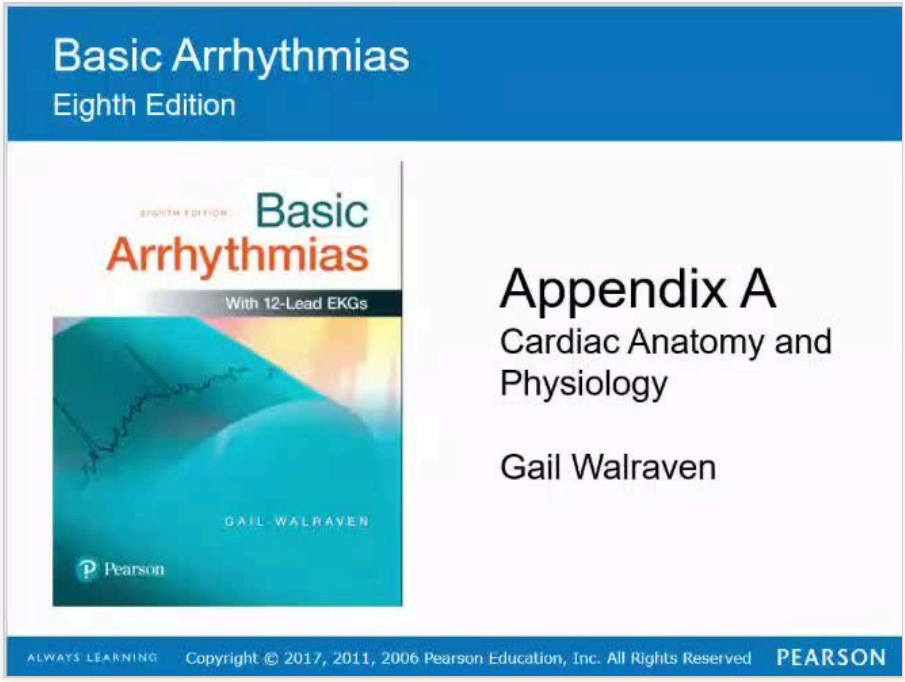
0:16:08 Coronary Circulation

0:19:53 Cardiac Cycle

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1 Appendix A: Cardiac Anatomy and Physiology (title slide)

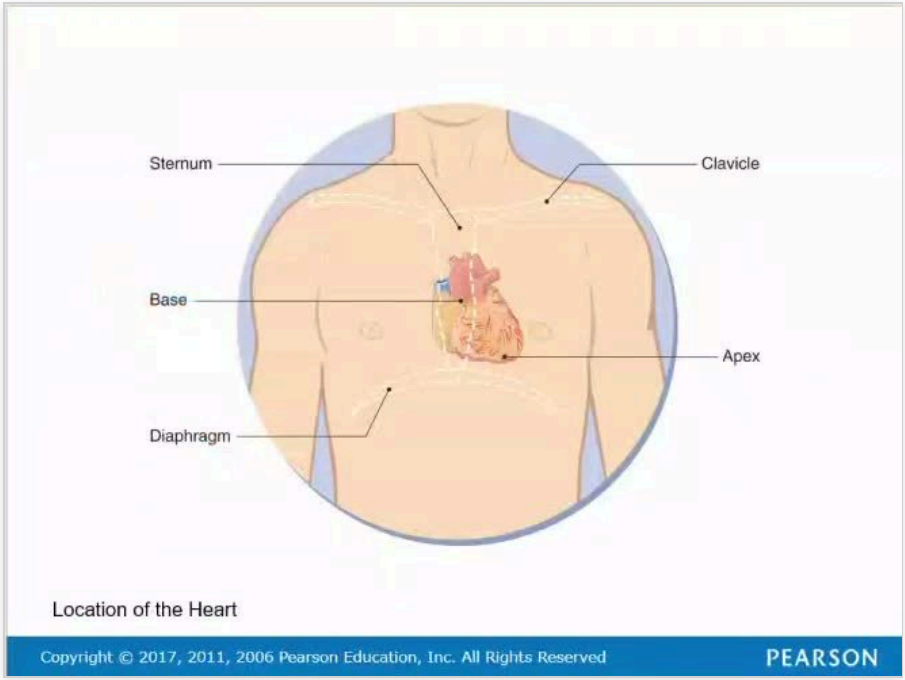
0:00:00 – 0:00:35



0:00:00 Welcome to the first lecture of Breath 101, Spring Semester, 2024. The first thing I want to talk about during this lecture is cardiac anatomy and physiology. So we're going to talk about where the heart sits. We're going to talk about the anatomy, all of the different portions of the heart itself. And we're going to talk about pumping or physiology of how the heart works normally. So as I go to the

2 Location of the Heart

0:00:35 – 0:01:23

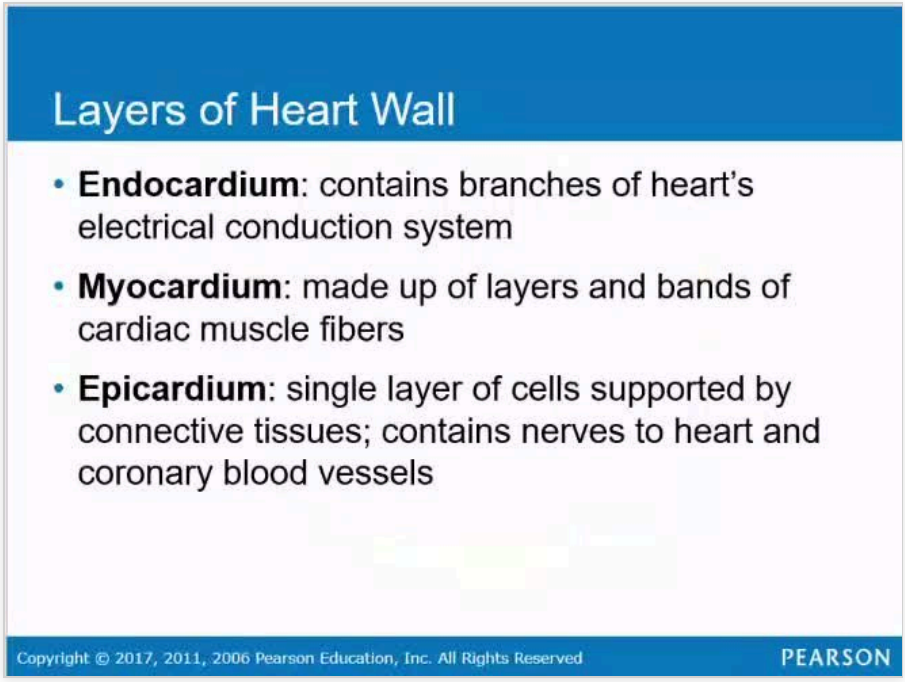


0:00:36 first slide here, we are going to discuss the position of the heart. So it sits in the middle of the chest, just slightly to the left. It is anterior in the chest, meaning it's more towards the front of the chest than it is towards the back of the chest. So it actually sits kind of right underneath the sternum or the breastbone.

0:00:57 And the position of the heart is important because when we place electrodes on to monitor somebody's cardiac rhythm or perform a 12-lead EKG, we need to make sure that those electrodes are placed appropriately around the heart surface so that we can get an accurate electrical reading.

3 Layers of Heart Wall

0:01:23 – 0:03:10

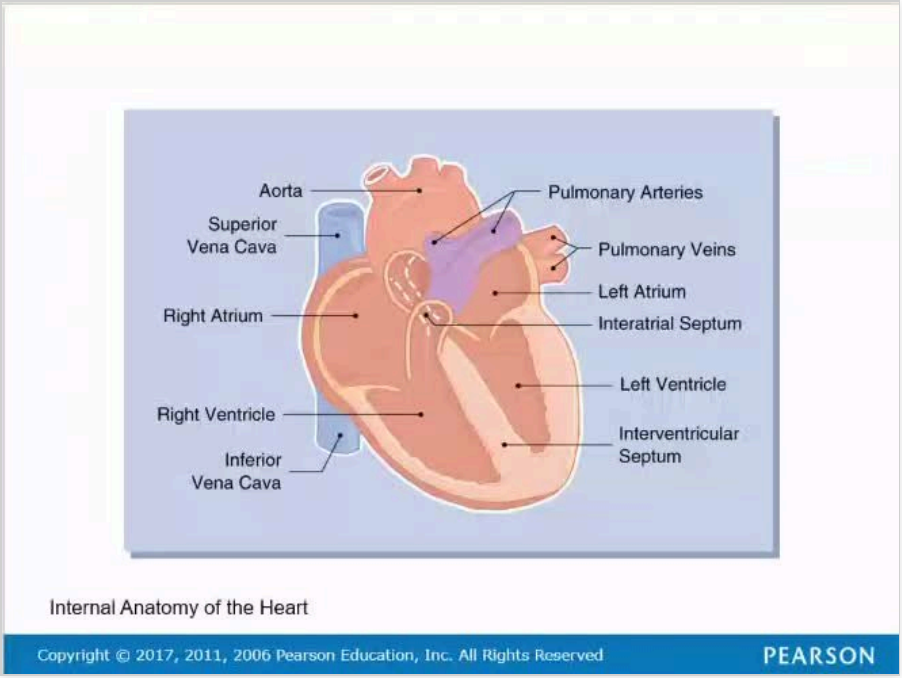


0:01:24 So I want to talk about the different layers of the heart wall. There's three layers on this slide, the endocardium. The endocardium is a layer that contains the heart's electrical conduction system. In other words, the nerves that carry the electrical impulse and that electrical impulse is delivered to the heart muscle cells in order for the heart muscle cells to contract. The middle layer is the myocardium and that is the layer and bands of cardiac muscle fibers. So that is the workhorse. Those are the cells that actually contract pushing blood out. The epicardium is a single layer on the outside of the heart which contains the nerves and also contains coronary blood vessels, also known as coronary arteries. These are the arteries that come off the base of the aorta and go around the outside of the heart to supply blood flow to the heart muscle cells. And this would be specifically where if somebody had coronary arteries disease, They have some kind of blockage in their coronary artery, which would

0:02:54 decrease blood flow to the heart muscle cells themselves and potentially cause what we call a heart attack, otherwise known as a myocardial infarction. We'll talk more about that as the semester goes on. So now we're

4 Internal Anatomy of the Heart

0:03:10 – 0:07:06



0:03:13 going to discuss blood flow through the heart. I like to be organized about this. It's important to understand it in an organized fashion because when somebody has an arrhythmia or a heart attack, it is going to affect the way that the heart beats and understanding the blood flow going through the heart will also help us understand the pathophysiology or how the heart will beat

0:03:46 abnormally during that heart attack or abnormal arrhythmia. So venous blood goes back to the right side of the heart via the superior vena cava and inferior vena cava. Those two large vessels empty blood into the upper chamber on the right side of the heart. The upper chambers are called the atrium or atria. In this case, it delivers venous blood or deoxygenated blood to the right atrium. When the right atrium contracts, it pushes blood through a valve, a valve that sits between the upper chamber and the lower chamber, and this specific valve is called the tricuspid valve on the right side. When blood goes to the tricuspid valve, it enters the right ventricle. When the right ventricle contracts, it pushes blood up through, there's a valve that sits on the bottom of the pulmonary artery and that is called the pulmonic valve or the pulmonary semilunar valve. Then blood enters the main pulmonary artery, also known as the pulmonary trunk, and then the pulmonary trunk splits into the right and left pulmonary

0:05:10 artery, bringing blood to the left and right lung. In the lungs, which is where gas exchange takes place, we get oxygen that diffuses from the lung into the blood and then carbon dioxide from the venous blood, which diffuses from the blood into the lung and then is exhaled. Then

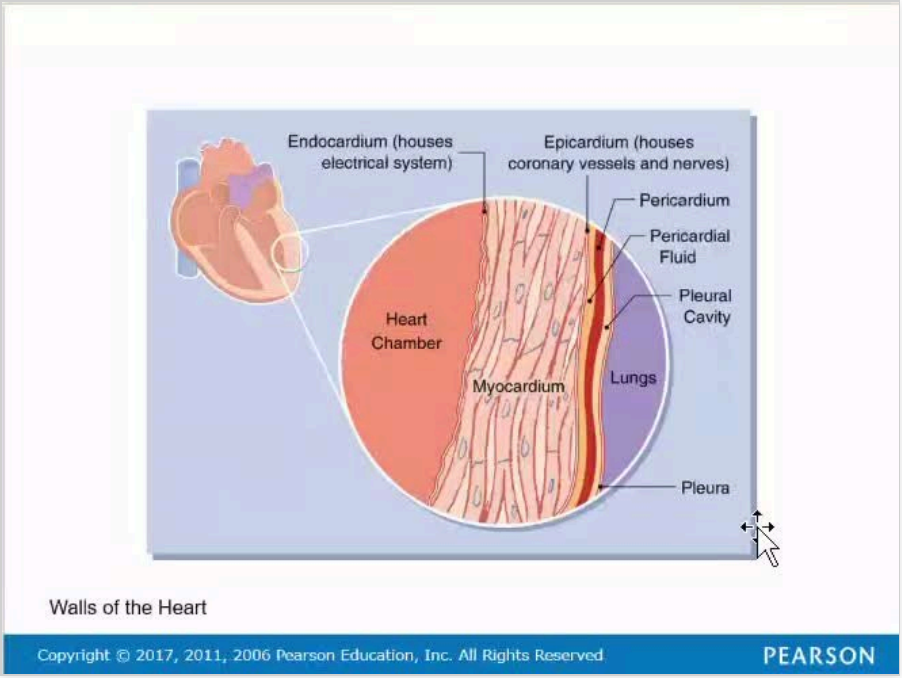
the blood continues on into the pulmonary veins, which now has oxygenated blood, bringing blood back to the upper chamber on the left-hand side, which is the left atrium.

0:05:53 When the left atrium contracts, it pushes blood through another valve that sits between the upper chamber and lower chamber on the left side. This valve is most commonly known as the mitral valve.

0:06:07 The other name for this valve would be the bicuspid valve. then blood enters the lower chamber or the left ventricle. When the left ventricle contracts, it pushes blood through a valve that sits at the base of the aorta aortic artery, also known as the aortic valve or the aortic semi-lunar valve. When blood goes through that valve, it goes into the ascending aorta, the aortic arch, and then the descending aorta where that oxygenated blood is distributed to the cells of the body for oxygen and nutrients where it picks up carbon dioxide and then goes back to the vein venous side of the blood flow and then comes back to the right side of the heart and that process starts all over again. Now we discussed this slide already just a small picture here on

5 Walls of the Heart

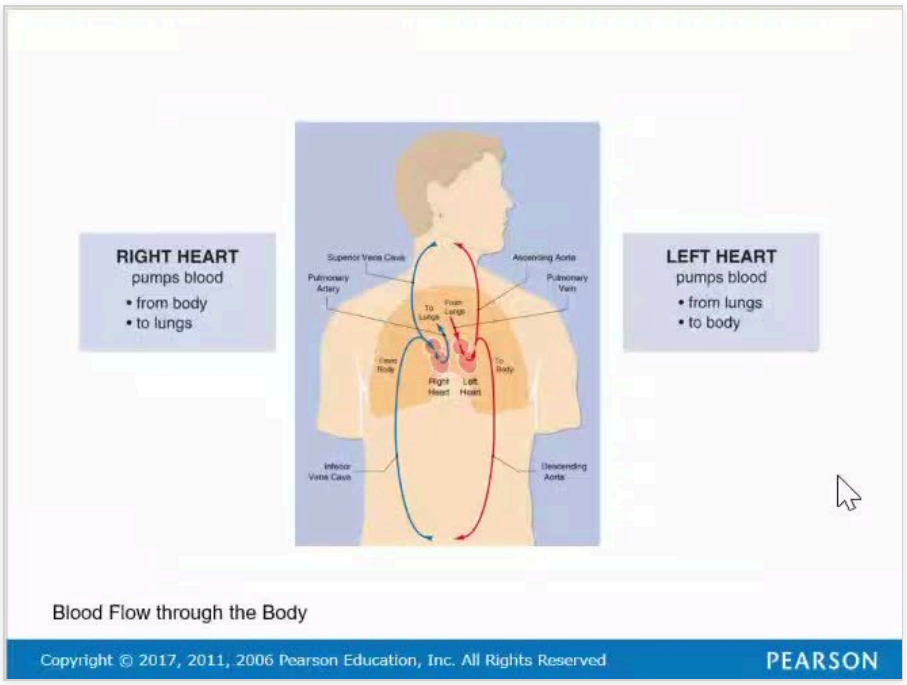
0:07:06 – 0:07:32



0:07:11 this slide showing the endocardium with the nerves, the myocardium is the muscle fibers, that's the workhorse that does all the contractions, and then the outer layer, which is the epicardium containing some nerves and blood vessels, the coronary artery blood vessels.

6 Blood Flow through the Body

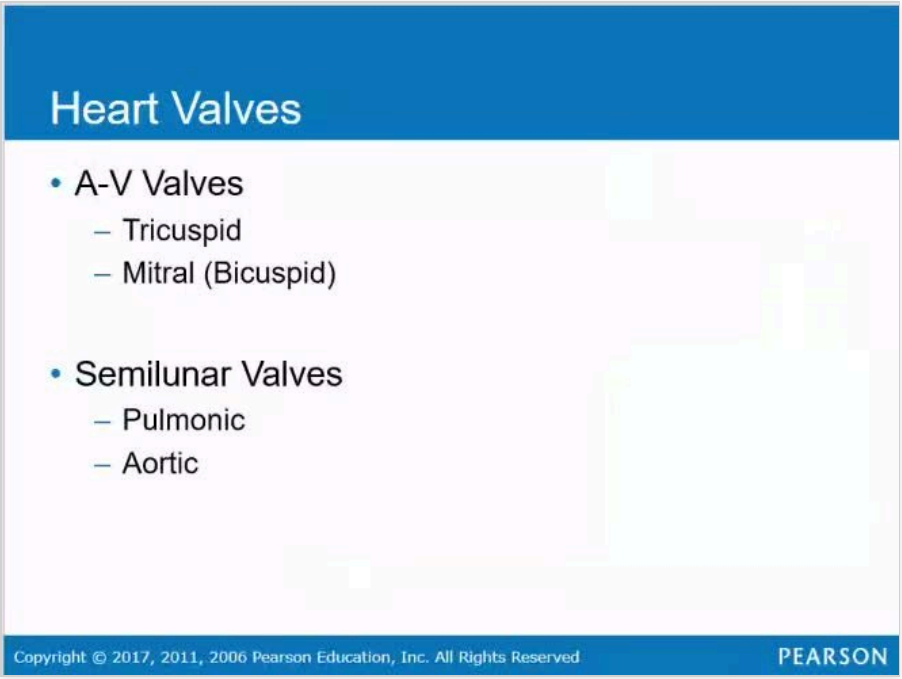
0:07:32 – 0:08:16



0:07:35 This slide is just showing that the left side of the heart beats oxygenated blood, which which is why the blood vessels are red, oxygenated blood out to the body and the cells, and then that is a high pressure system. And the right side of the heart, which mainly contains deoxygenated blood, which is why the vessels are blue, is really a low pressure system. You don't need a lot of pressure to pump blood through the lungs to the right side, but you do need a lot of pressure to pump the blood all the way out to all of the cells and the distal portions of your hands and feet.

7 Heart Valves

0:08:16 – 0:10:42



0:08:17 OK, we talked about these valves briefly. We have what's called AV valves. AV stands for atrioventricular valve. So the two upper chambers are called atria. Two lower chambers are called ventricular or ventricles. So these valves sit right between the upper and lower chambers as the term AV valves. The one that's on the right side is the tricuspid valve, the one that's on the left side is the mitral valve. And then I talked about the semi-lunar valves. These semi-lunar valves sit at the bottom of the arteries where the right ventricle pushes it through the pulmonic valve into the pulmonary artery and the left ventricle pushes it through the aortic valve into the aortic artery.

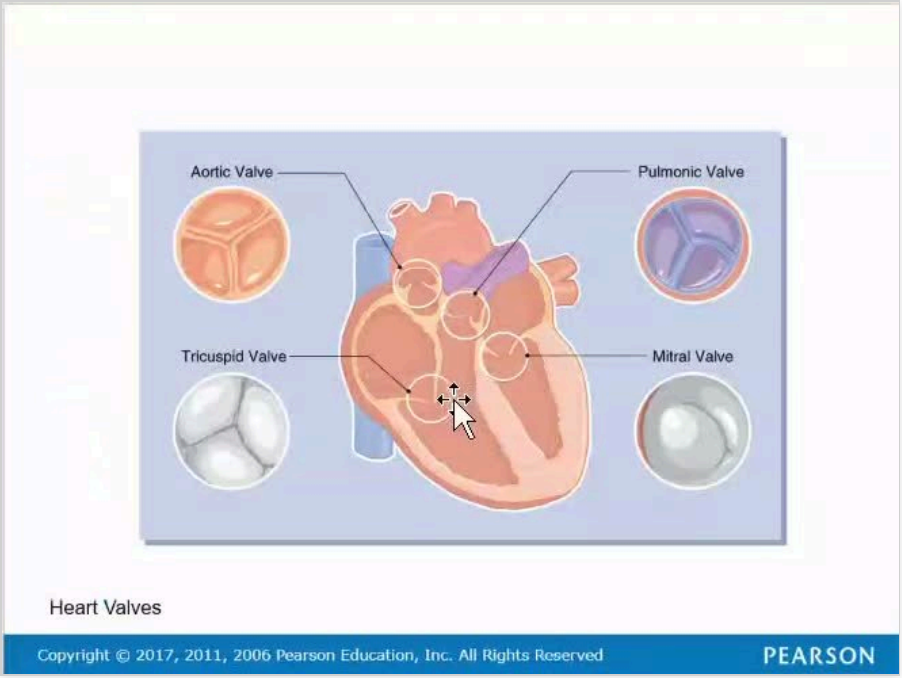
0:09:12 These valves can sometimes present a problem if they don't function correctly. There's basically two types of valve dysfunction. A stenotic valve, also known as stenosis, where the valve doesn't open all the way, and when it doesn't open all the way, it impedes blood flow moving forward. So you have less blood flow or restricted amount of blood flow moving forward. The other problem that these valves can present is something called regurgitation. I said earlier that the valves are one-way valves that allow blood to go forward but not back. A regurgitated valve doesn't close all the way, so when a lower chamber contracts blood flow, some blood flow can actually go backwards instead of going forward.

0:10:10 Why I mention this is because when we do 12 lead EKGs, this valve problem can cause an electrical variance and we can pick that up on a 12 lead EKG.

0:10:28 It can also cause problems in the way that the heart beats and if left untreated for long periods of time can really affect somebody's right ventricle or left ventricle and its function. So as I said the AV valve sit between the lower

8 Heart Valves (diagram)

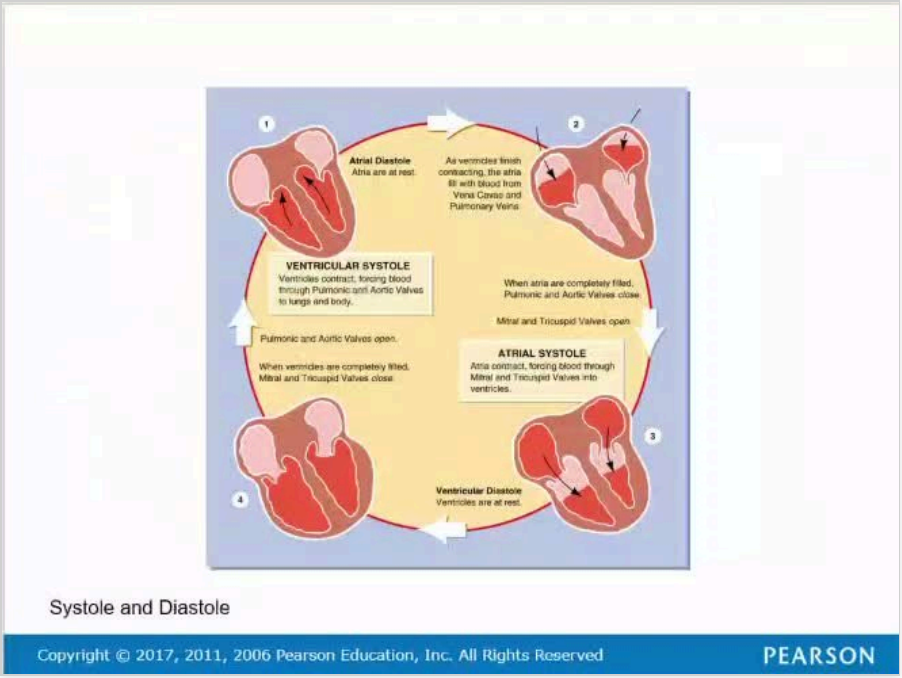
0:10:42 – 0:10:59



0:10:47 upper and lower chambers, tricuspid on the right, mitral on the left, and then the pulmonic valves at the base of the arteries. So I'm not going to go in

9 Systole and Diastole

0:10:59 – 0:16:08



0:11:10 direct order on this slide here. So I'm actually gonna start at number two on the slide. This slide is important to understand how the heart best functions.

0:11:29 The heart best functions when it beats from the top down. Oh, I wanna talk about atrial filling, atrial contraction first. So as blood comes back to the right atrium via the IVC SVC and blood comes back to the left atrium via the pulmonary veins and fills the right and left atria. The AV valves during atrial filling are slightly open so as the atria fill some blood actually filters down into the right ventricle and left ventricle. So these AV valves are not closed during atrial filling.

0:12:25 When the atria are filled, the atrial will now contract, moving on to number three on this slide, push all of its blood content down into the ventricles.

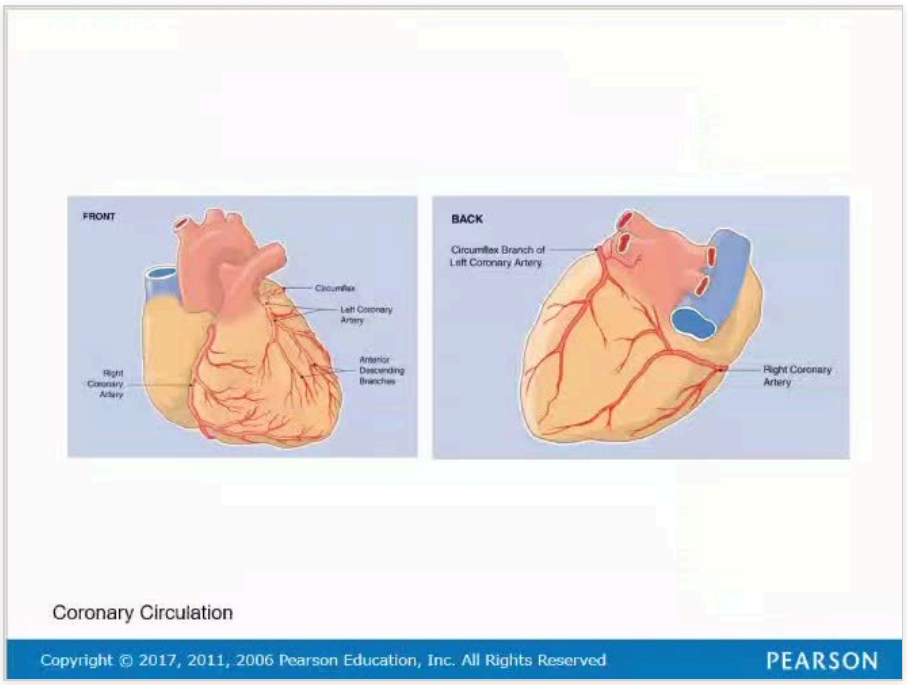
0:12:41 And that's what I mean from the top down. We want the atria to contract and push all of its blood content down into the ventricles. We want to maximize the amount of blood down in the ventricle. The reason for that is because when the ventricles contract, in a normal heart, the more blood it has there, the more blood gets ejected. The amount of blood that gets ejected per beat is known as stropholium. So we want the heart to be from the top down so that we can maximize stroke volume. And you can see up here is in number four is maximized ventricular filling. And then number one is where we get ventricular contraction, pushing blood out into the pulmonary artery and out into the aorta.

0:13:37 Now the reason why it's important, this understanding is important because the amount of blood that the heart beats per minute is called cardiac output. There's two components to that. The number of times that the heart beats per minute, otherwise known as the heart rate, and the amount of blood ejected per beat, otherwise known as stroke volume. So the formula for cardiac output is stroke volume times heart rate. And cardiac output is best supported by stroke volume and not heart rate. If you think about in terms of heart rate, the faster the heart contracts, the less time it has to fill with blood.

0:14:24 So if there's less blood in the chamber when it contracts, your stroke volume gets smaller. So when we have really fast heart rates, it can affect your cardiac output because your stroke volume will drop. So other terms that are important on this slide here is you see some terminology of atrial diastole, atrial systole, ventricular diastole, and ventricular systole. So let's break down these terms. Diosdole means that the chamber is relaxed and is filling with blood. So diastole is relaxation or filling. The cardiac muscle cells are also repolarizing and resetting itself so that it can accept and respond to the next stimulus. So when I say diastole I'm referring to relaxation or repolarization. Systole means that the chamber is actually contracting or I could say depolarization. So when I say systole contraction depolarization, it all means the same thing. So atrial systole means atrial contraction or atrial depolarization. So it's important to understand those terms. As I move on to the next slide, here is where you're

10 Coronary Circulation

0:16:08 – 0:19:53



0:16:11 going to see coronary arteries that are on the outside of the heart. You can see that the root of the aorta over here, and then we have the right coronary artery and then the left coronary artery coming there. So the right coronary artery eventually branches out and provides oxygenated blood to basically the right side of the heart muscle cells and the left coronary artery, which also branches out and usually one artery from this side will branch around the back that's called the circumflex and it will feed the back side of the heart muscle cells and then we have the anterior portion of that left coronary artery feeding the front side of the left side of the heart and then you can see the back we have the circumflex coming around and then the right coronary artery also branches around back feeding the posterior sides of the left and right side. What's important to understand about that is if we get some kind of blockage in one of the coronary arteries and it decreases blood

0:17:31 flow, so if we had a blockage in the right coronary artery way up high, it would block blood flow to all of the cardiac muscle cells on the right side of the heart. And what happens when we don't have oxygen or your cells don't get oxygen. They get ischemic because they don't have enough oxygen to support cellular metabolism. If that blockage lasts for a longer period of time, the cells can then become injured and if the blockage still continues, then the cells can die. So this is the process of what is called the heart attack or myocardial infarction. What's important to understand here is, you know, as I said earlier, if somebody's having a heart attack, they might come in complaining of shortness of breath, or chest pain, or

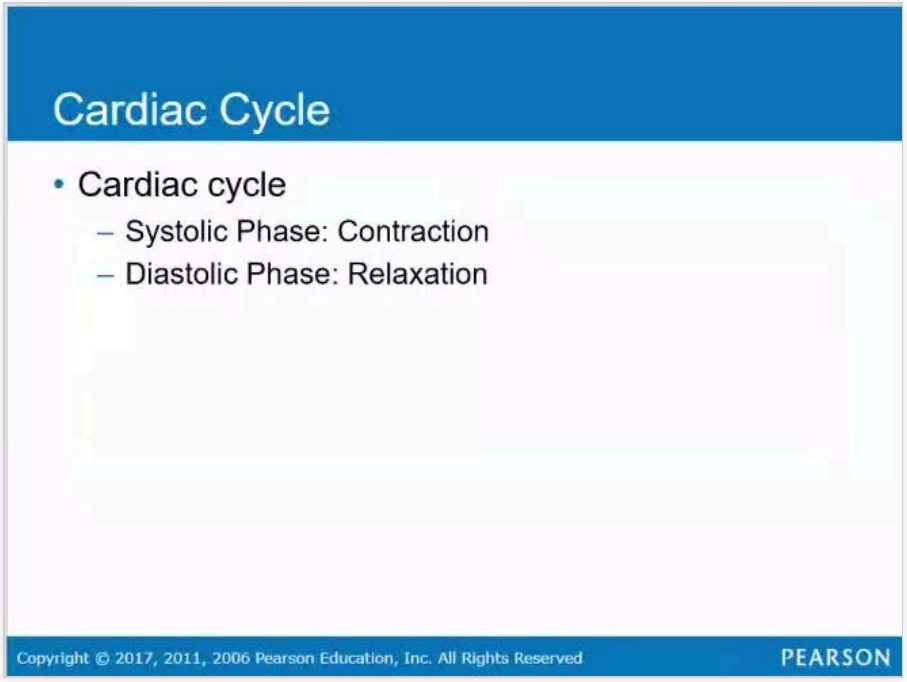
lightheadedness, or jaw pain, arm pain, leg pain, radiating back pain. One of the first things we would do for that patient is we would think, hmm, this patient having a heart attack, we'll put them on

0:18:45 some oxygen and we'll get a 12-lead EKG. Where we place electrodes on the chest and the arms and legs, we get an electrical recording of the heart. And then if somebody's having a heart attack, there will be an electrical abnormalities that show up on the 12-lead EKG. We get 12 different electrical views and we would look at each one of those leads and see where the electrical abnormality appears, depending on what lead it is, we know what electrode that portion of the heart is covering, so then we can kind of narrow down what coronary artery is causing the problem. And then we would either send them to the cath lab, or they can go through a cardiac cath, or they inject dye to see where the blockage is.

0:19:37 If there's enough blockage and they can fix it, they might put a coronary artery spint in, or if the blockage is really bad, they would send them to the OR for open heart surgery.

11 Cardiac Cycle

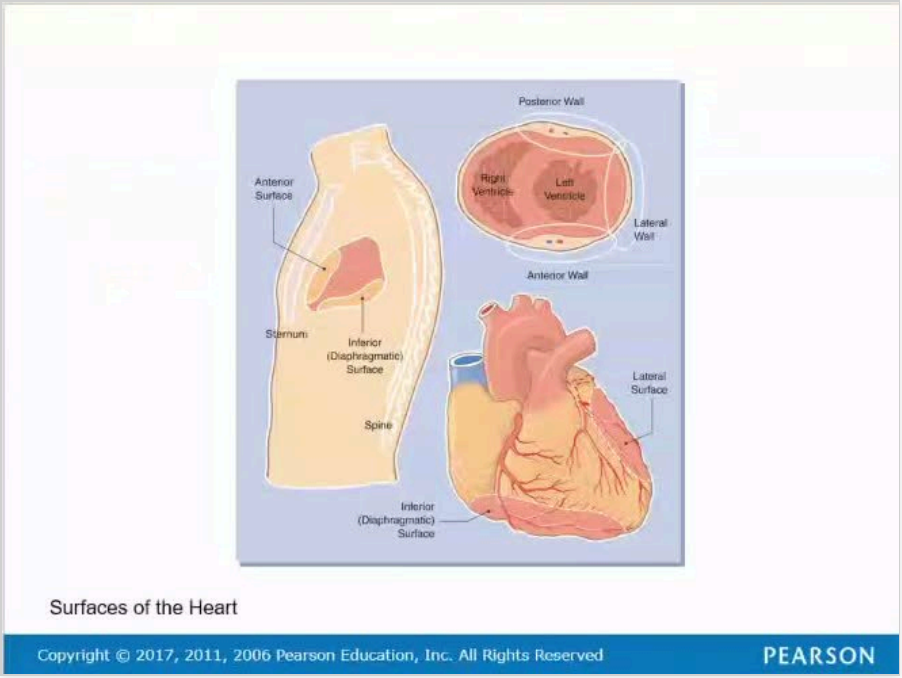
0:19:53 – 0:20:16



0:19:54 So I talked about this already, that the terminology systolic phase or systole means contraction, also means depolarization. Diastole phase means relaxation or repolarization. those terms are important, we'll reuse them throughout the semester. And then as I was

12 Surfaces of the Heart

0:20:16 – 0:21:51



0:20:18 talking about the heart attack a few seconds ago when we do a 12 lead EKG, whatever lead the electrical abnormality shows up, we know what surface of the heart it is. And there's several surfaces. If we're looking at the heart from an anterior section, there's a lower wall here, which is known as the inferior wall. There's a wall on the left side of the left side of the heart known as the lateral surface, and then there's the anterior wall. So the anterior wall would be here, and so if somebody has electrical abnormality and a lead that reflects the anterior section, right, we know it's what arteries will supply that anterior wall of the heart and it's usually the left coronary artery more specifically the left anterior descending coronary artery. There's also a posterior wall in the back so the back side of the heart somebody could have a posterior wall MI and then there's also a band of cardiac muscle cells that sit between the right and the left side of

0:21:34 heart that's called the septal wall. Somebody can also have a septal wall MR. And that concludes our lecture on anatomy and physiology of the heart.